ITI – RedHat Administration I Lab 1

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1. –

```
Loading...

Welcome to Fedora 33 (riscv64)

[root@localhost ~]# pwd
/root
[root@localhost ~]# adduser mohammadtarek
[root@localhost ~]# passwd mohammadtarek
Changing password for user mohammadtarek.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# su - mohammadtarek
[mohammadtarek@localhost ~]$ pwd
/home/mohammadtarek
[mohammadtarek@localhost ~]$
```

2. **What is the difference between cat and more command?**
   a. cat prints the entire content of a file to the standard output **only one time**
   b. more prints the content batch by batch and allow the user to transfer from the current batch to next
3. **What is the difference between rm and rmdir using man?**
   a. rm removes files
   b. rmdir removes directories
4. _Create the following hierarchy under your home directory

```
[mohammadtarek@localhost ~]$ mkdir dir1
[mohammadtarek@localhost ~]$ mkdir docs
[mohammadtarek@localhost ~]$ cd dir1
[mohammadtarek@localhost dir1]$ mkdir dir11
[mohammadtarek@localhost dir1]$ mkdir dir12
[mohammadtarek@localhost dir1]$ cd dir11
[mohammadtarek@localhost dir11]$ touch file1
[mohammadtarek@localhost dir11]$ cd ../..
[mohammadtarek@localhost ~]$ cd docs
[mohammadtarek@localhost docs]$ touch mycv
[mohammadtarek@localhost docs]$
```

a. Could’nt remove non-empty dir using ‘rmdir’, so used ‘rm -r ‘

```
[mohammadtarek@localhost ~]$ cd dir1
[mohammadtarek@localhost dir1]$ rmdir dir11
rmdir: failed to remove 'dir11': Directory not empty
[mohammadtarek@localhost dir1]$ rm -r dir11
[mohammadtarek@localhost dir1]$
```

b.dir1 also deleted..

```
[mohammadtarek@localhost dir1]$ cd ..
[mohammadtarek@localhost ~]$ rmdir -p dir1/dir12
[mohammadtarek@localhost ~]$ pwd
/home/mohammadtarek
[mohammadtarek@localhost ~]$ ls
docs
[mohammadtarek@localhost ~]$
```

c.

```
[mohammadtarek@localhost ~]$ pwd
/home/mohammadtarek
[mohammadtarek@localhost ~]$ /home/mohammadtarek/docs/mycv
-bash: /home/mohammadtarek/docs/mycv: Permission denied
[mohammadtarek@localhost ~]$ cd /home/mohammadtarek/docs/mycv
-bash: cd: /home/mohammadtarek/docs/mycv: Not a directory
[mohammadtarek@localhost ~]$ cat /home/mohammadtarek/docs/mycv
[mohammadtarek@localhost ~]$ cat docs/mycv
[mohammadtarek@localhost ~]$
```

5. _Copy the /etc/passwd file to your home directory making its name is mypasswd.

```
[mohammadtarek@localhost ~]$ rm mypasswd
[mohammadtarek@localhost ~]$ ls
docs
[mohammadtarek@localhost ~]$ cp /etc/passwd /home/mohammadtarek/
[mohammadtarek@localhost ~]$ ls
docs  passwd
[mohammadtarek@localhost ~]$ mv passwd mypasswd
[mohammadtarek@localhost ~]$ ls
docs  mypasswd
[mohammadtarek@localhost ~]$
```

6. _Rename this new file to be oldpasswd.

```
[mohammadtarek@localhost ~]$ mv mypasswd oldpasswd
[mohammadtarek@localhost ~]$ ls
docs  oldpasswd
[mohammadtarek@localhost ~]$
```

7. _You are in /usr/bin, list four ways to go to your home directory (cd $home)

```
[mohammadtarek@localhost ~]$ cd /usr/bin
[mohammadtarek@localhost bin]$ pwd
/usr/bin
[mohammadtarek@localhost bin]$ cd
[mohammadtarek@localhost ~]$ cd /usr/bin
[mohammadtarek@localhost bin]$ cd ~
[mohammadtarek@localhost ~]$ cd /usr/bin
[mohammadtarek@localhost bin]$ cd /home/mohammadtarek
[mohammadtarek@localhost ~]$
```

8. _
9. _Display the first 4 lines of /etc/passwd

```
[mohammadtarek@localhost ~]$ head -n 4 /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
[mohammadtarek@localhost ~]$
```

10. _Display the last 7 lines of /etc/passwd

```
[mohammadtarek@localhost ~]$ tail -n 7 /etc/passwd
geoclue:x:990:990:User for geoclue:/var/lib/geoclue:/sbin/nologin
qemu:x:107:107:qemu user:/:/sbin/nologin
kojibuilder:x:989:988::/builddir:/bin/bash
chrony:x:988:987::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
pesign:x:987:986:Group for the pesign signing daemon:/var/run/pesign:/sbin/nologin
mohammadtarek:x:1000:1000::/home/mohammadtarek:/bin/bash
[mohammadtarek@localhost ~]$
```

11. _Display the man pages of passwd the command and the file sequentially in one command.

```
[mohammadtarek@localhost ~]$ man passwd $$ man 5 passwd
PASSWD(1)                                User utilities                                PASSWD(1)

NAME
       passwd - update user's authentication tokens

SYNOPSIS
       passwd [-k] [-l] [-u [-f]] [-d] [-e] [-n mindays] [-x maxdays] [-w warndays] [-i inactivedays] [-S] [--stdin] [-?] [--usage] [username]

DESCRIPTION
       The passwd utility is used to update user's authentication token(s).

       This  task  is achieved through calls to the Linux-PAM and Libuser API.  Essentially, it initializes itself as a "passwd" service with Linux-PAM
       and utilizes configured password modules to authenticate and then update a user's password.

       A simple entry in the global Linux-PAM configuration file for this service would be:

        #
        # passwd service entry that does strength checking of
        # a proposed password before updating it.
        #
        passwd password requisite pam_cracklib.so retry=3
        passwd password required pam_unix.so use_authtok
        #

       Note, other module types are not required for this application to function correctly.

OPTIONS
       -k, --keep-tokens
              The option -k is used to indicate that the update should only be for expired authentication tokens (passwords); the user wishes  to  keep
```

12. Display the man page of the passwd file.

```
[mohammadtarek@localhost ~]$ man 5 passwrd
No manual entry for passwrd in section 5
[mohammadtarek@localhost ~]$ man 5 passwd
PASSWD(5)                                Linux Programmer's Manual                                PASSWD(5)

NAME
       passwd - password file

DESCRIPTION
       The  /etc/passwd  file  is  a text file that describes user login accounts for the system.  It should have read permission allowed for all users
       (many utilities, like ls(1) use it to map user IDs to usernames), but write access only for the superuser.

       In the good old days there was no great problem with this general read permission.  Everybody could read the encrypted passwords, but the  hard-
       ware  was  too slow to crack a well-chosen password, and moreover the basic assumption used to be that of a friendly user-community.  These days
       many people run some version of the shadow password suite, where /etc/passwd has an 'x' character in the password field, and the encrypted pass-
       words are in /etc/shadow, which is readable by the superuser only.

       If  the  encrypted  password, whether in /etc/passwd or in /etc/shadow, is an empty string, login is allowed without even asking for a password.
       Note that this functionality may be intentionally disabled in applications, or configurable (for example using the "nullok"  or  "nonull"  argu-
       ments to pam_unix.so).

       If the encrypted password in /etc/passwd is "*NP*" (without the quotes), the shadow record should be obtained from an NIS+ server.

       Regardless  of  whether  shadow  passwords are used, many system administrators use an asterisk (*) in the encrypted password field to make sure
       that this user can not authenticate themself using a password.  (But see NOTES below.)

       If you create a new login, first put an asterisk (*) in the password field, then use passwd(1) to set it.

       Each line of the file describes a single user, and contains seven colon-separated fields:

           name:password:UID:GID:GECOS:directory:shell
```

13. _Display a list of all the commands that contain the keyword passwd in their man page.

```
[mohammadtarek@localhost ~]$ man -k passwd
passwd: nothing appropriate.
[mohammadtarek@localhost ~]$
```